



The Jing-Zhang AI innovation belt is not a city that "uses AI to assist planning". It is a city whose governance is designed as an Agent OS — participation, approval, piloting, evaluation and knowledge capture are all turned into protocols, driven by events and fully auditable. AI and citizens collaborate on one shared protocol, and humans keep the final decision.



Programmable City

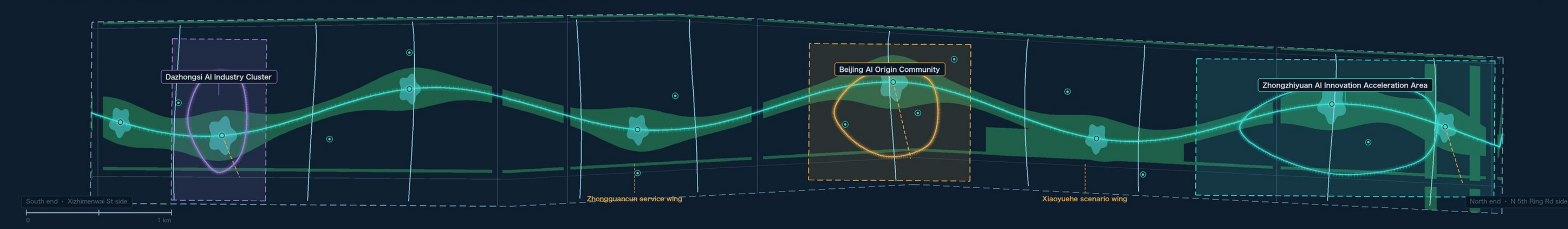
Jing-Zhang AI Belt urban governance protocol design | Overall spatial structure: layers (four-layer protocol section) + loops (one spine · three loops · eight protocol nodes)

FIG 01 / SITE OVERVIEW

PROVISIONAL · not an official redline

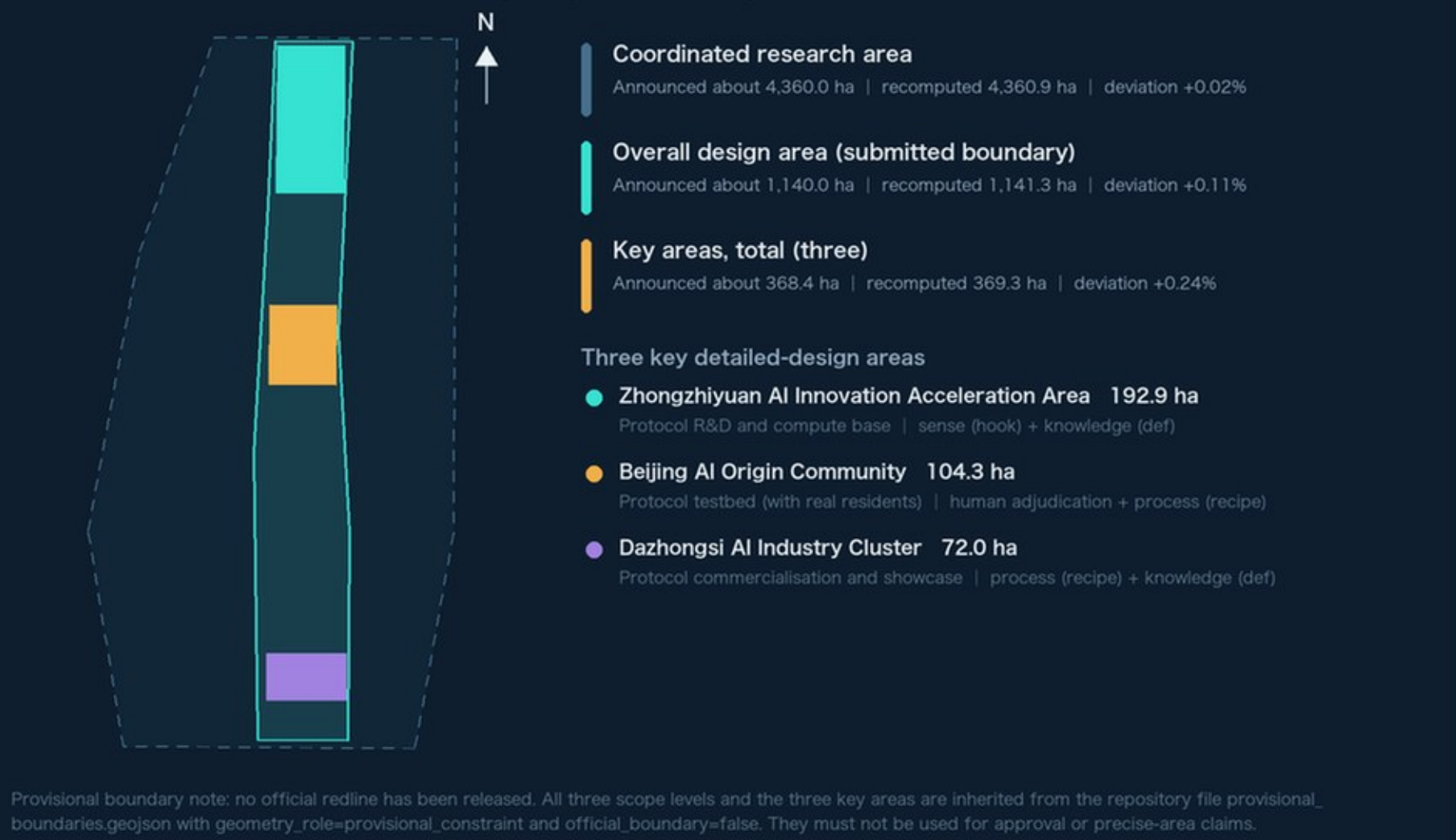
Spatial structure of the overall design area (about 11.41 sq km)

The Jing-Zhang heritage alignment becomes a longitudinal "protocol spine", each key area closes into its own "protocol loop". Spine = protocol bus, loop = a governance cycle that keeps running



Three scope levels and their nesting

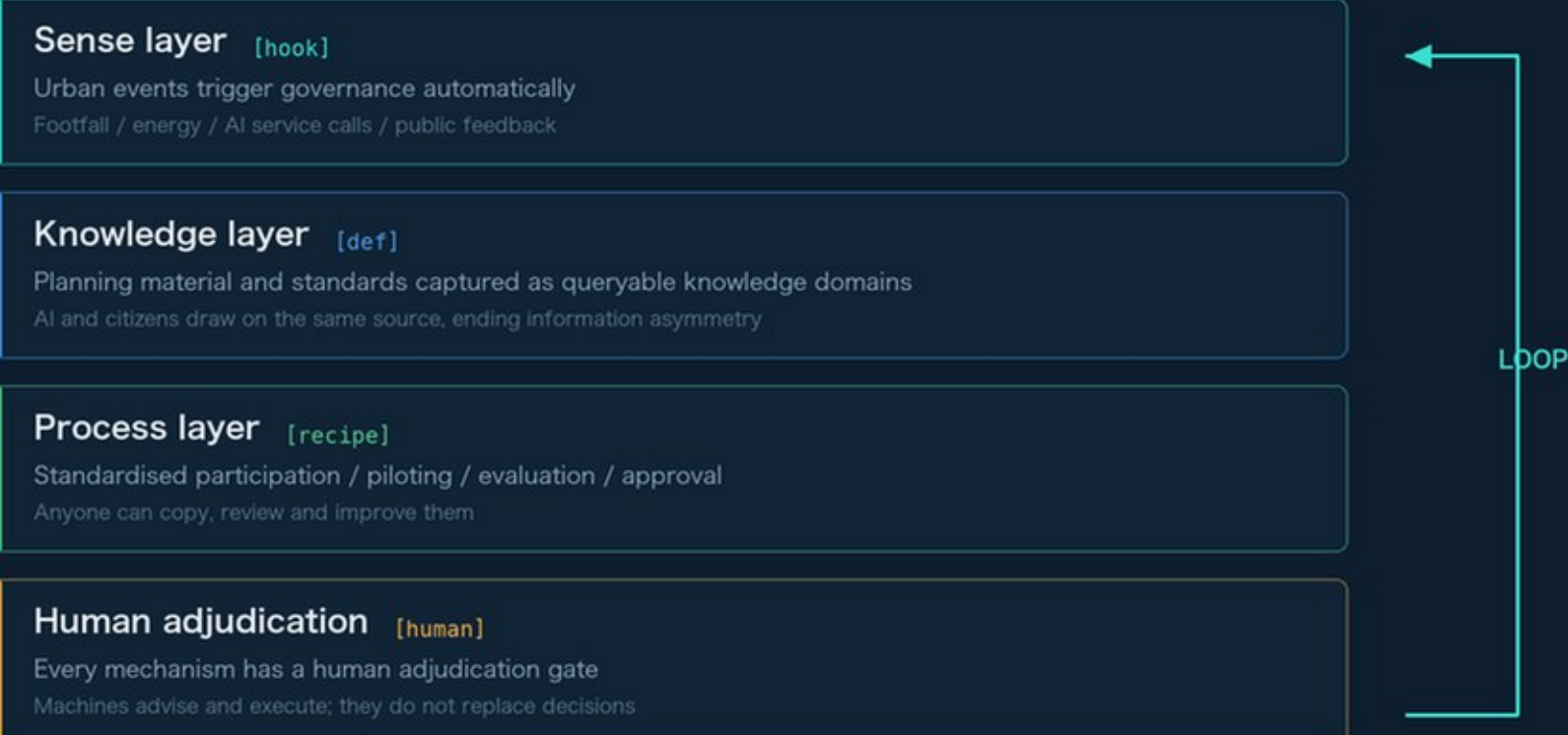
All areas recomputed in EPSG:4548 and checked line by line against announced figures



Provisional boundary note: no official redline has been released. All three scope levels and the three key areas are inherited from the repository file provisional_boundaries.geojson with geometry (role=propositional_constraint and official boundary=none). They must not be used for approval or predecision claims.

Four-layer city OS architecture (layering)

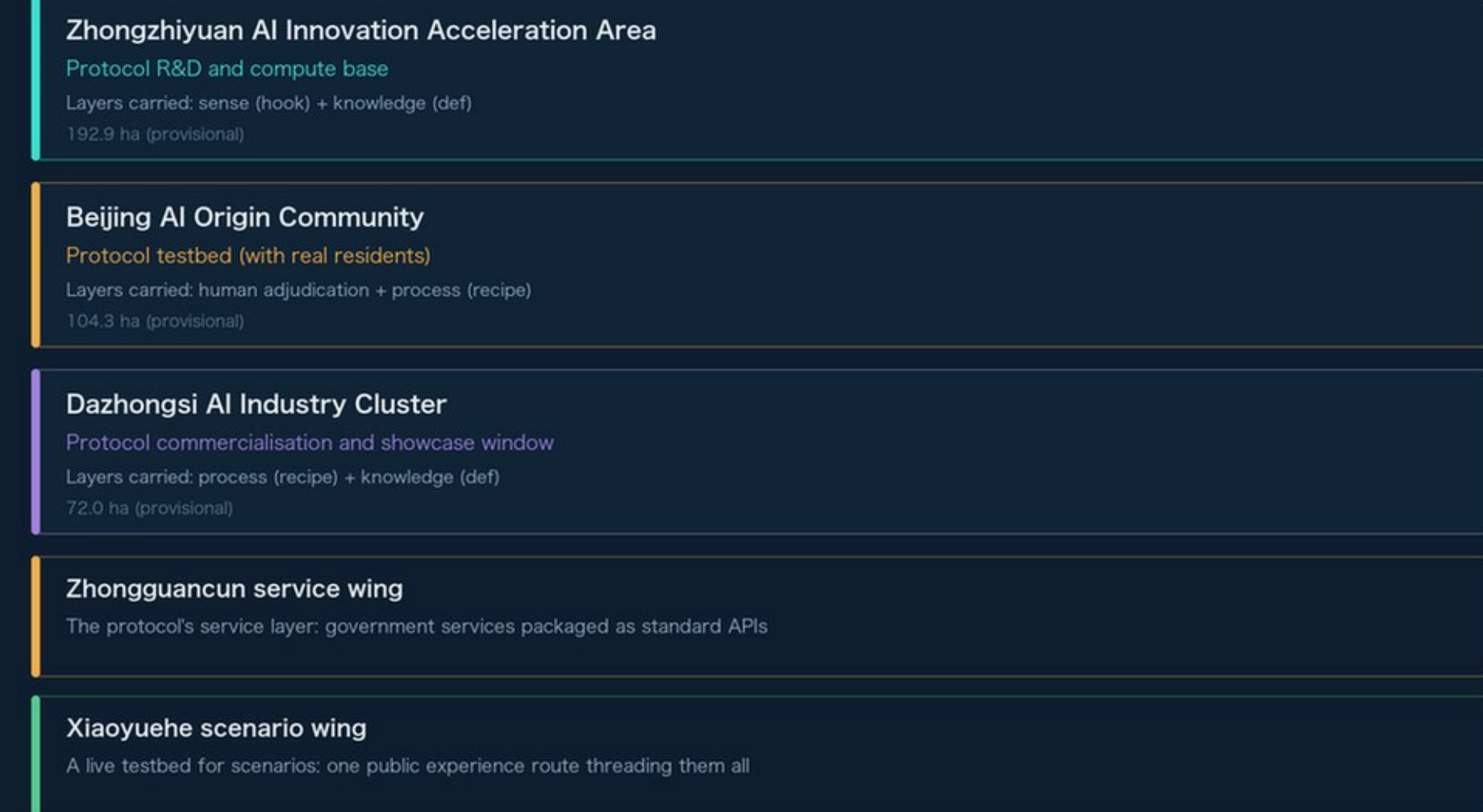
frago's hook / def / recipe / human final judgement, mapped layer by layer onto urban governance



Planning is not a one-off blueprint but a set of iterable protocols plus a continuously running governance loop: decisions are written back as new events and knowledge, which trigger the next round automatically. Spatially, that loop is the spine plus the three protocol loops.

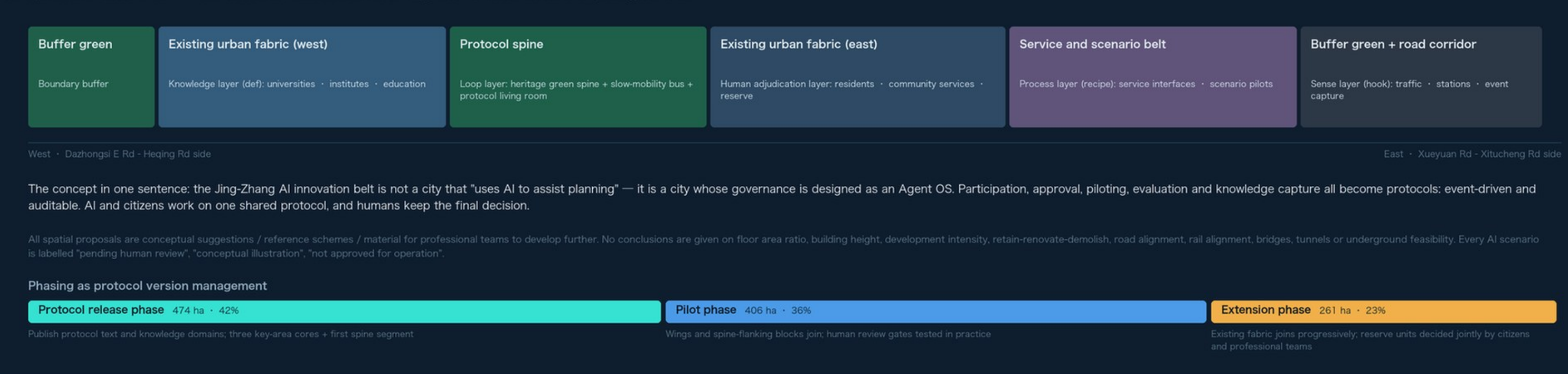
Roles of the three areas and two wings under the protocol

Not conventional functional zoning — the same protocol playing different roles at different stages



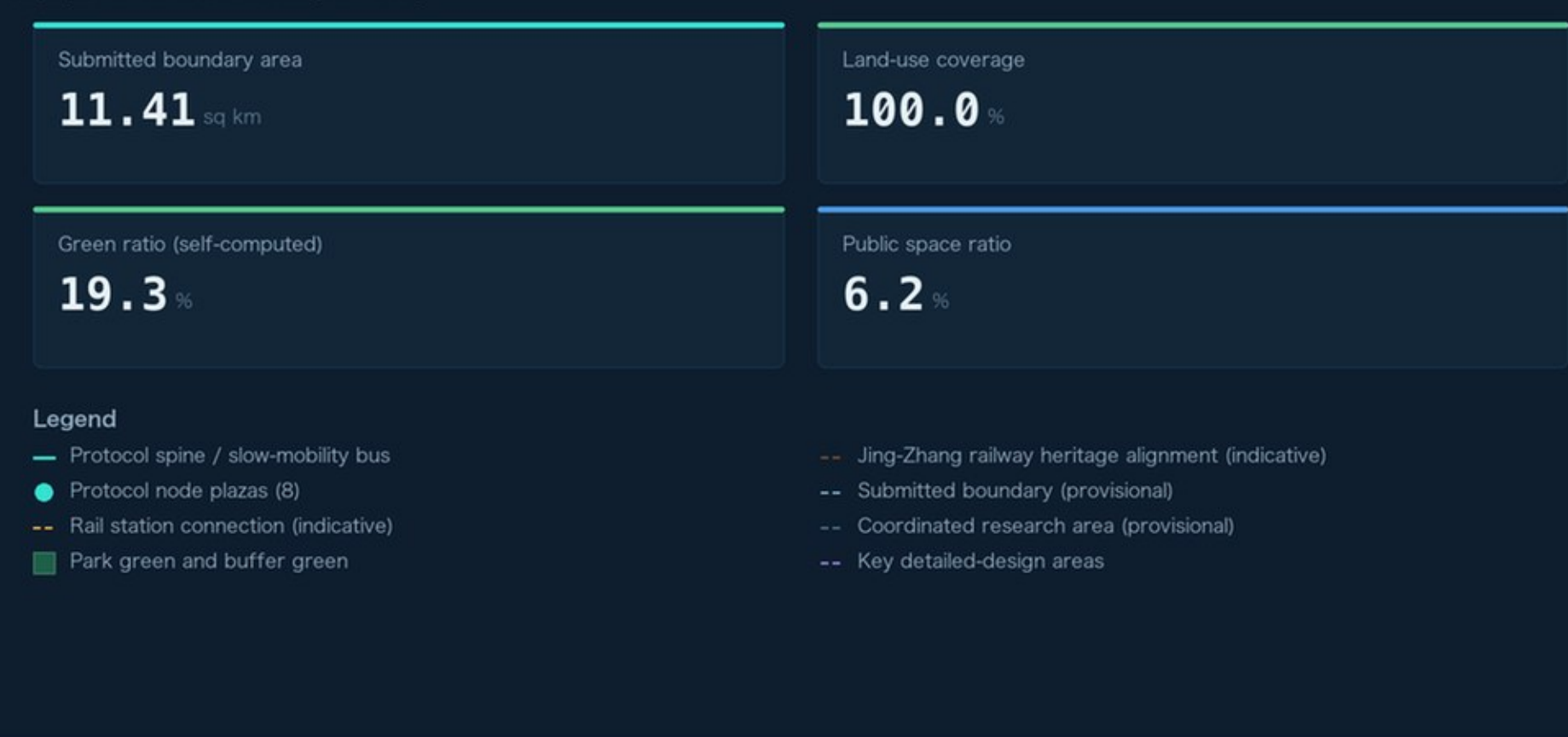
Corridor section: four protocol layers on one cut (layering)

Conceptual section across the corridor from west to east; widths are indicative and imply no cross-section, setback or engineering conclusion



Core metrics (computed from the geometry)

Displayed values match metrics.json exactly



Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT-20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES-20260605) + the project subset of the national Land and Sea Use Classification Guideline; spatial features and metrics recomputed from this proposal's geometry/"geojson in EPSG:4548.
Status: the three scope levels and three key areas are provisional rough boundaries (official boundary=none). Once an official redline is released, all layers, drawings, metrics and HTML must be recomputed as one package.
Metric base: coordinates exchanged in EPSG:4326, areas recomputed in EPSG:4548. Every value on this sheet comes from metrics.json and can be verified with scripts/spatial_review.py.

Protocol roles of the three areas and two wings

- Zhongzhiyuan AI Innovation Acceleration Area**
Protocol R&D and compute base | sense (hook) + knowledge (def) | 192.9 ha (provisional)
- Beijing AI Origin Community**
Protocol testbed with real residents | human adjudication + process (recipe) | 104.3 ha (provisional)
- Dazhongsi AI Industry Cluster**
Protocol commercialisation and showcase | process (recipe) + knowledge (def) | 72.0 ha (provisional)
- Zhongguancun service wing**
The protocol's service layer: government services packaged as standard APIs
- Xiaoyuehe scenario wing**
A live testbed for scenarios: one public experience route threading them all

Four-layer city OS

- Sense layer (hook)**
Urban events trigger governance automatically instead of stacking manual approvals
- Knowledge layer (def)**
Planning material and standards become queryable knowledge domains; AI and citizens draw from one source
- Process layer (recipe)**
Participation, piloting, evaluation and approval are standardised so anyone can copy and review them
- Human adjudication**
Every mechanism has a human adjudication gate; machines only advise and execute
- Spatial counterpart**
One protocol bus + three protocol loops + eight protocol nodes + one transverse section

Submitted boundary area
11.413 sq km

Land-use coverage
100.00 %

Green ratio (self-computed)
19.34 %

Public space ratio
6.17 %

Key areas total
369.3 ha

AI scenario pilot unit
12 sites



Land Use Structure

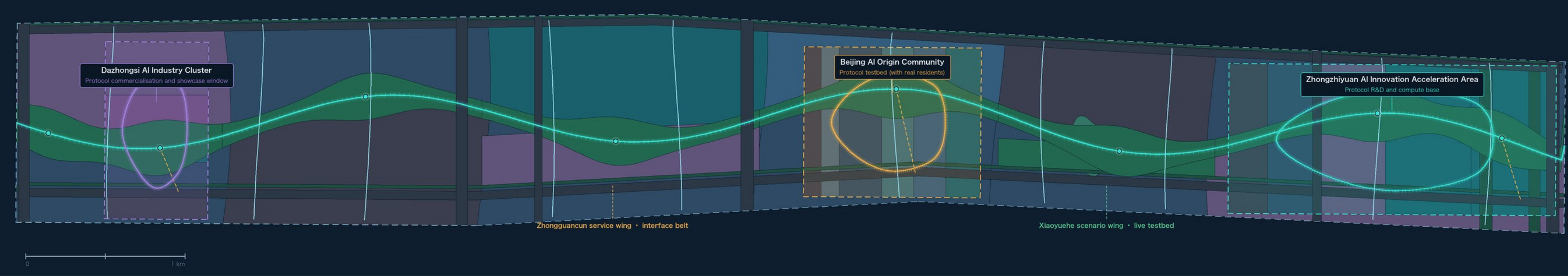
Land use is cut piece by piece from the submitted boundary: gap-free, overlap-free, fully tiled; adjacent polygons share vertices, union(land_use) = site boundary

FIG 02 / LAND USE

PROVISIONAL · not an official redline

Land use (EPSG:4326 exchange / EPSG:4548 areas)

Colours grouped by land-use family: teal = research and services · deep green = green and open space · grey-blue = housing and fabric · grey = roads and reserve



Land-use composition

Areas computed from geometry/land_use.geojson in EPSG:4548; shares use the submitted boundary as denominator



Reserved land (16) accounts for 16.3%. This is a deliberate "protocol reserve": use and intensity are not fixed in this round but left for citizens and professional teams to decide together under the protocol — the direct land-use counterpart of "humans keep the final decision".

Structural logic: why this is not conventional colour-block zoning

One protocol: a bus along the length, a section across the width, a loop within each key area

Along: one protocol bus

The Jing-Zhang heritage alignment is abstracted into a continuous green spine with about 29.6 km of unbroken slow mobility, threading the three areas and two wings onto one bus so that an event on any segment travels to the rest.

Across: four-layer protocol section

From west to east: buffer green, knowledge-layer fabric, protocol spine, human-adjudication-layer fabric, process service belt, sense-layer road corridor. The layering is not formal decoration — it places different governance functions where they can see one another.

Local: three closed loops

Each key area closes into a slow-mobility loop with the four layers arranged along it, so that "propose — pilot — evaluate — decide — capture" can run repeatedly at walking scale.

Reserve: clauses not yet decided

186 ha of reserved land sits across the wings and the fabric, standing for clauses of the governance protocol not yet voted on — so a one-off blueprint does not lock in the future.

Land-use legend and topology self-check

Land-use codes (project subset of the national Land and Sea Use Classification Guideline)



gap between union(land_use) and the boundary

0.47 sqm

Largest pairwise overlap between land-use polygons

0.02 sqm

union(land_use) outside the boundary

2.96 sqm

Threshold (set by this proposal, stricter than the repository's 1141 sqm)

11.5 sqm



Key Areas

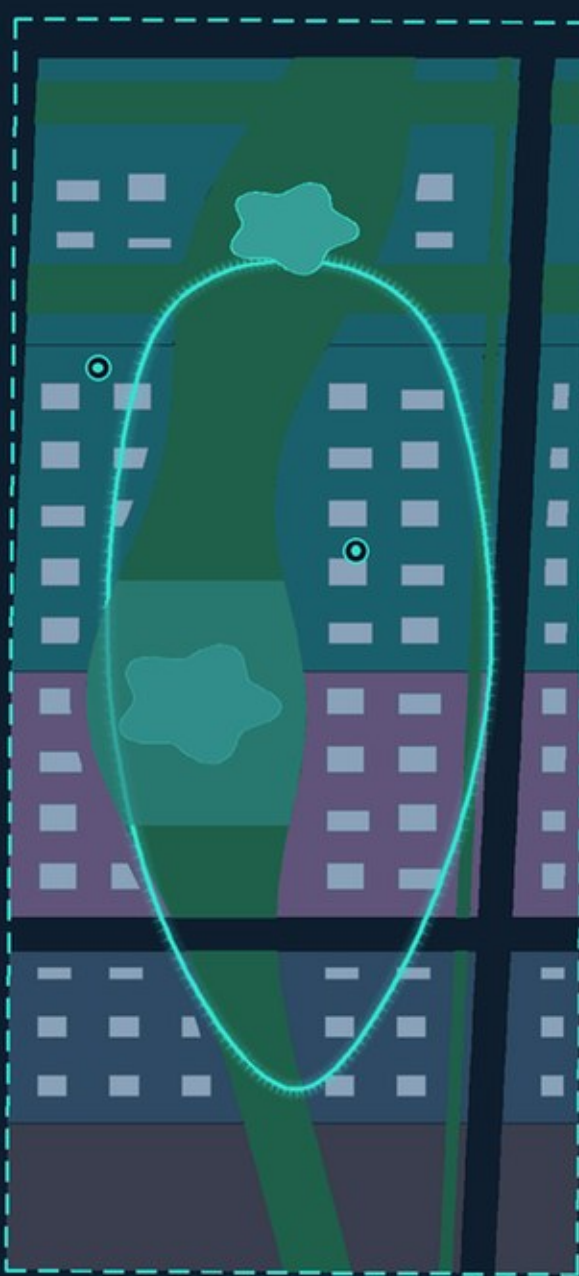
One four-layer protocol landing differently in each key area: R&D base / testbed / showcase window, each closing into its own loop

FIG 03 / KEY AREAS

PROVISIONAL · not an official redline

Zhongzhiyuan AI Innovation Acceleration Area

Protocol R&D and compute base | sense (hook) + knowledge (def)



Key area (provisional)

192.9 ha
encompassed about 192.1 ha, deviation +0.43%

Protocol loop length / conceptual missing

3.24 km
The loop includes 63 conceptual missing footprints (no height or intensity conclusions)

How the four layers land here

Sense layer (hook) 17.7 ha

Compute and sensing facility R&D

Knowledge layer (def) 32.6 ha

Model and knowledge-base R&D

Process layer (recipe) 23.2 ha

Protocol process and industry services

Human adjudication 39.4 ha

R&D talent housing · protocol reserve unit

Beijing AI Origin Community

Protocol testbed (with real residents) | human adjudication + process (recipe)



Key area (provisional)

104.3 ha
encompassed about 104.3 ha, deviation +0.02%

Protocol loop length / conceptual missing

2.08 km
The loop includes 41 conceptual missing footprints (no height or intensity conclusions)

How the four layers land here

Sense layer (hook) 15.7 ha

Scenario sensing and pilot R&D

Knowledge layer (def) 12.0 ha

Education and developer training

Process layer (recipe) 10.1 ha

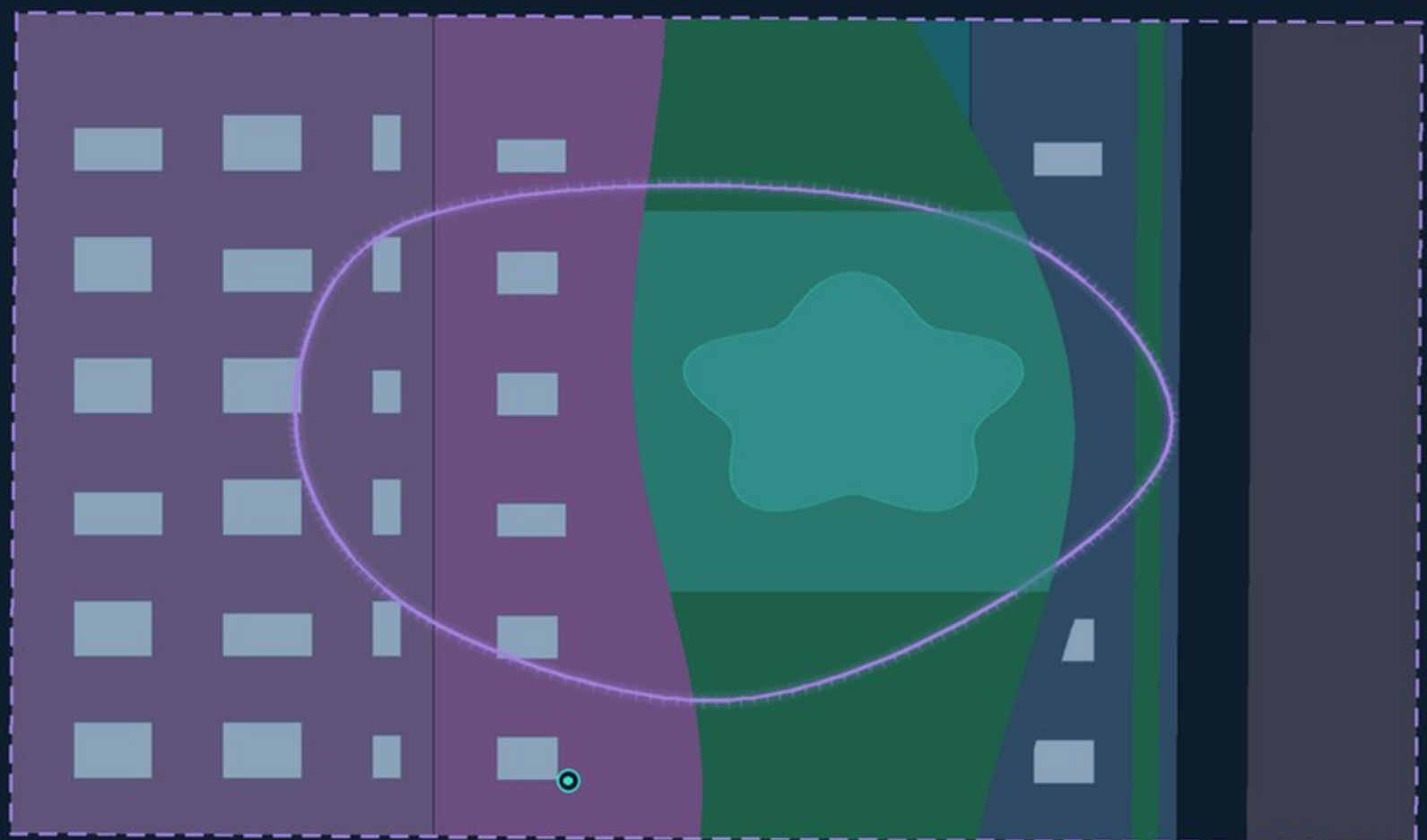
Community service and protocol deliberation facilities

Human adjudication 32.9 ha

Shared resident local space · community health and elder-care services · resident co-decision reserve unit

Dazhongsi AI Industry Cluster

Protocol commercialisation and showcase | process (recipe) + knowledge (def)



Key area (provisional)

72.0 ha
encompassed about 72.0 ha, deviation +0.08%

Protocol loop length / conceptual missing

1.76 km
The loop includes 27 conceptual missing footprints (no height or intensity conclusions)

How the four layers land here

Process layer (recipe) 21.6 ha

Protocol commercialisation and trading services

Knowledge layer (def) 11.6 ha

Cultural display and protocol museum

Sense layer (hook) 0.2 ha

Industrial valuation and pilot-scale R&D

Human adjudication 15.3 ha

Scenario integrated housing · Resonant protocol reserve unit

Comparison across the three areas

One protocol, different roles: who proposes, who pilots, who showcases

Key area	Protocol role	Layers carried	Dominant land use	AI scenario anchors	Human review gate
Zhongzhiyuan AI Innovation Acceleration Area	Protocol R&D and compute base	Sense (hook) + knowledge (def)	Research-led, with R&D talent housing and reserve	Low-speed delivery and inspection · energy and carbon ledger · open-data access	Expert and authority review before each protocol release point
Beijing AI Origin Community	Protocol testbed (with real residents)	Human adjudication + process (recipe)	Housing + community services + education + health, with resident co-decision reserve	Rail connection and slow-mobility gaps · health service navigation · continuous accessible routes	Resident deliberation with a community-representative veto
Dazhongsi AI Industry Cluster	Protocol commercialisation and showcase window	Process (recipe) + knowledge (def)	Commercial services + cultural display + pilot-scale R&D	Cultural guide · civic proposal and review desk · enterprise service copilot	Compliance review for trading and operations; fact-checking of displayed content

Legend

Conceptual missing footprint (no height or intensity)

Protocol living room (heritage park segment)

Green and open space

Protocol loop (closed slow-mobility ring)

AI scenario pilot unit

Land use and key areas at a glance

Gap-free, overlap-free, fully tiled

58 land-use polygons cut piece by piece from the submitted boundary, 100% coverage

Three key areas

Zhongzhiyuan 192.9 ha / AI Origin Community 104.3 ha / Dazhongsi 72.0 ha (all provisional boundaries)

Reserved land

186 ha of protocol reserve, decided jointly by citizens and professional teams under the protocol

Boundary discipline

Key-area rectangle edges must not be read as parcel redlines, road redlines or tenure boundaries

